



CELL EXTRACTION FROM STAINING IMAGES

SEMESTER PROJECT REPORT

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Abstract

Cell extraction from stained images is an intensely studied subject with a history of more than 50 years, but there still is not an absolute solution for all kinds of staining methods that encompasses the needs of multiple areas in biology. Also, the information that needs to be extracted from any segmentation process is ever-increasing. Therefore, it is hard to create methods which can be used for multiple different tissues without the need of too many parameter adjustments, but it is very necessary especially for cancer detection studies.

To extract the desired regions from any image, the technique of staining and necessary degree of segmentation is needed to be known. This report presents an extensive algorithm that can be applied to nearly every type of animal tissue staining using some very basic assumptions of cell shapes, and compares it with some state of the art methods. This algorithm starts by converting the given image to a 2D distance map with darker target regions, and applies a series of k-means and watershed algorithms to separate background from the desired regions. Then, with the help of morphological filters, a recursive convex cutting algorithm is applied to the regions with lower quality, according to a proposed score. The proposed method is able to work at an extensive array of staining methods with the calibration of one parameter and also outputs a probabilistic cell quality scoring which can be used to apply further region segmentation algorithms.

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Chapter 1

Introduction and Motivation

Cell extraction has several major challenges that require attention before the creation of a fully-functioning algorithm for different kinds of staining techniques. First of all, the conversion of color images to gray scale is a necessary step for nearly every segmentation method, but there is not a well-accepted method to complete this step without losing critical information.

Secondly, cell extraction algorithm needs to cover for both the intensity and shape of the target cells, which are parameters that may drastically change even in a single image for tissues with possible anomalies. As a final step, researchers need to be careful about dividing falsely segmented clusters.

For the former issue, the approaches are quite straightforward for many cell extraction and image analysis methods: From each channel R, G and B, a weighted average is calculated for each pixel and thus grayscale image is formed. In the situations where the user does not have enough knowledge about the composition of the images, these weights are usually taken as equal, a method studied under the name of “Intensity” by Kanan and Garrison in their paper. In different staining types, however, different weights can be used according to the target structures. This method works very well contrary to its simplicity, but depending on the staining technique, some other color space such as LAB or HSV may be used. For these cases, luminosity and value channels are usually extracted, respectively (Kanan, Garrison 2; Pinidiyaarachchi). The

lack of a method without user intervention or some necessary parameters, however, is still an ongoing problem.

As for the second challenge, there were several segmentation methods used throughout the history, and many of them consists of several steps at which various corrective algorithms are applied to the output of main segmentation process. These additional methods include region merging/splitting, morphological operations (Xiaoqiang 433), edge detectors, watershed and random walker algorithms (Xiaoqiang 434-435), region growing (Bengtsson 4-8), active contours (Shah 431) and many more (Meijering 1-6; Pinidiyaarachchi). These are generally applied after basic preprocessing and thresholding methods, to be able to extract data from a more probable structure.

After the main segmentation, the falsely segmented clusters issue comes to front as algorithms are usually unable to segment too close, over/under stained, damaged or multiplying cells. At this point, the use of convexity defects is a frequently utilized approach, along with ellipse fitting (Indumathi et al.; Kothari et al.; Wang et al.; Wienert et al.; Xing et al.).

1.1 State of the Art Algorithms

The first uses of computers for the analysis of cells date back more than half a century. Already in the mid-1950s, systems were developed to automate the classification of smears of exfoliated cells, with the aim to enable mass screening for cervical cancer by applying thresholding-based decision rules to serial one-dimensional (1D) microscopic line scans of a specimen (Meijering 1). Thresholding and feature targeting methods become the main algorithms to extract cell images for the next several decades, until the morphology and region

based methods started to take their places with the advancing technological possibilities in both biology and medical imaging (2-6).

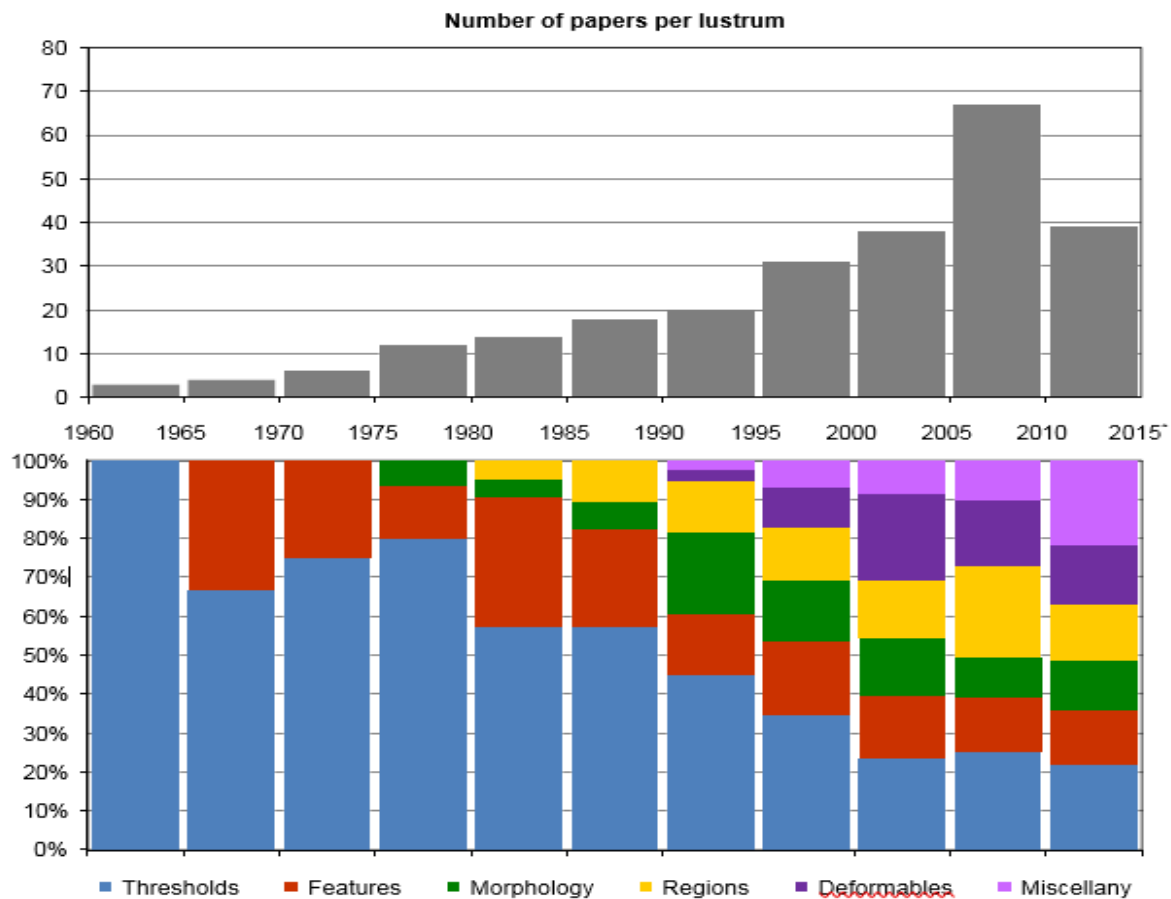


Figure 1 - Count of published papers and the distribution of the methods of cell extraction (Meijering 6)

With the development of machine learning and deep learning tools, modern-day methods started to utilize a combination of segmentation and detection techniques, where the latter is mainly based on a neural network, an SVM or a Bayesian classifier (Meijering; Xing and Yang). Main disadvantage of these approaches is the occasional need of a training set, which is usually acquired by the efforts of several researchers. For the segmentation methods, the trend is using a watershed or region growing algorithm: Out of 45 algorithms studied on the paper of

Xing and Yang, 28 of them uses a segmentation method and out of these, 15 of them is either a watershed algorithm, or a direct derivation (59-65).

The problem for the comparison of these different algorithms is the issue of different datasets: Since many of the papers written in the recent years use their own dataset, it is difficult to determine whether one approach is better than another. In addition, since different detection and segmentation measurements are used in the literatures, it is not straightforward to quantitatively compare the current methods (Xing and Yang 36).

1.2 Structure of the Proposal

In this work, in addition to proposing a nuclei segmentation method for a wide array of staining types, there is also a scoring system that can be used to compare different methodologies for the same image sets, without the use of any baseline detection method. This scoring will also be used to enhance the abilities of preprocessing, K-means/watershed segmentation and cell cutting methods that will be presented in this report.

Chapter 2

Methodology

2.1 Color Image Preprocessing

As large part of the staining and imaging techniques produce a color image, first objective of this work is to convert color images into grayscale ones, with a resulting image that is darker (i.e. has a lower intensity value) on the target regions. For this purpose, K-means clustering with $K=3$ is applied to the three-layered image, to divide it into theorized areas of background, cytoplasm and nuclei.

The assumption at this point is the circularity and relatively small size of the target regions:

After the K-means, every K layer is individually labeled and a score is created out of them, as seen below:

Cell Score = Area / Circularity, where

Circularity = $4\pi \times \text{Area} / \text{Perimeter}^2$, thus creating:

Cell Score = $(\text{Perimeter}^2 / 4\pi) \propto \text{Perimeter}^2$

Cell Score = Perimeter²

This simple value lets us to choose the center that is connected to the most blobs, which is the desired input. By calculating the mean cell score of each region in the label, ideally for the ones

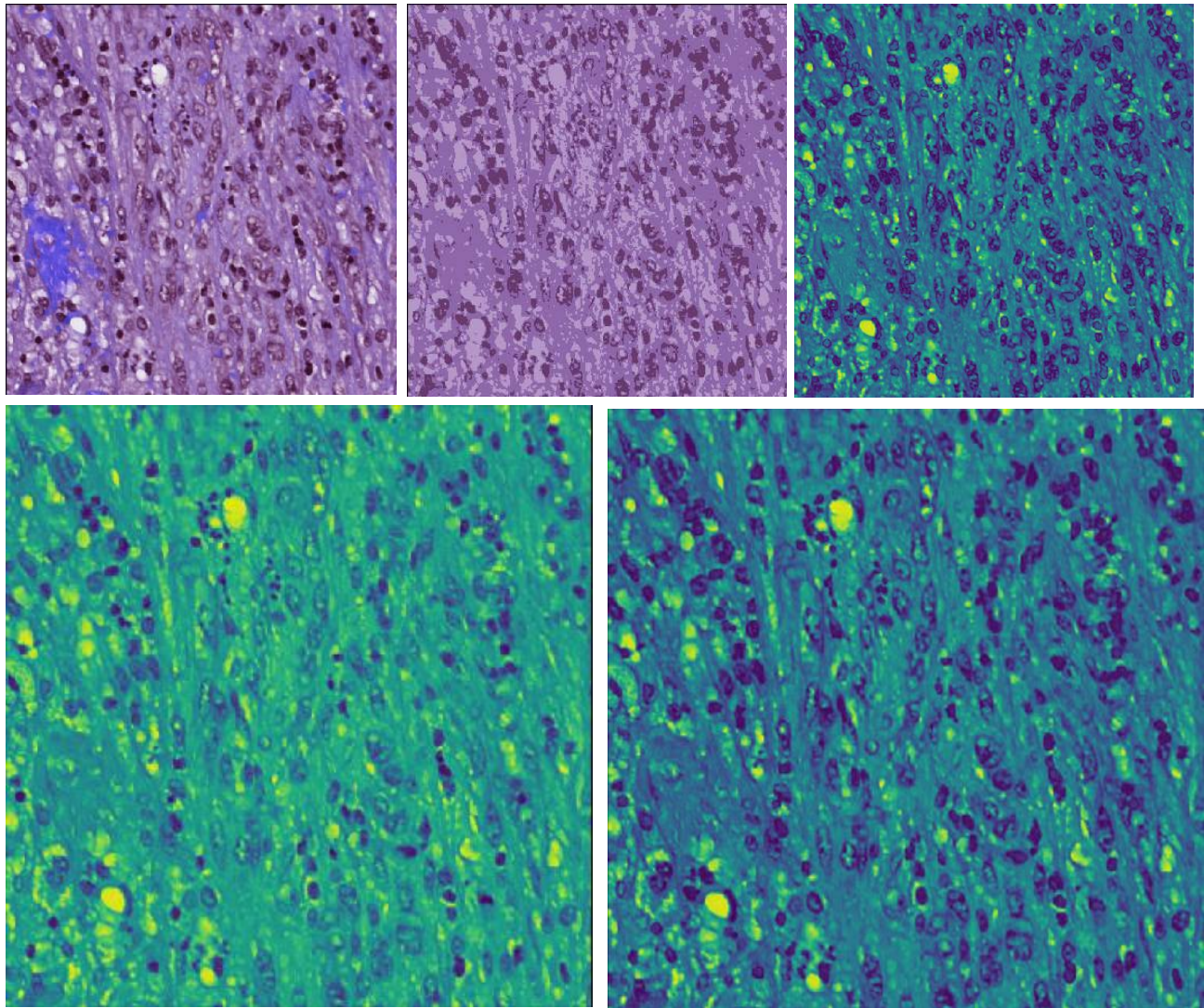


Figure 2-6 - From left to right, top to bottom: Initial image, K-means segmentation for K=3, Distance map of non-normalized center, Current state of the art <Intensity> method, Proposed method of normalized distance map

that are larger than a certain pixel value (64 is taken in our cases), and extracting the center that has the lowest cell score, the probable color of the desired regions is found.

Before creating a distance map to this color, an additional process is necessary as

this center is not ideal to determine the nuclei centers which should behave as the minimal values of their neighbors. This center, however, is ideal to use as an edge detector, an effect that can be observed at the Figure 4.

To capture the minima inside the border of the cell, we need to apply a contrast normalization to the given center value: Since the distance of the extracted center to other centers should be higher as the channel gets more descriptive (and vice versa), subtracting the difference of the

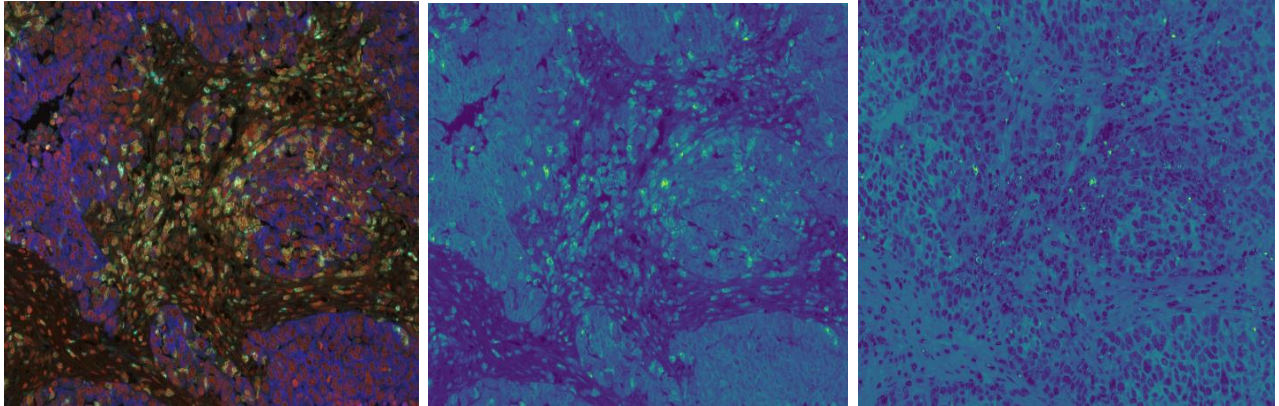


Figure 7-9 - From left to right: Initial image with multicolored staining, <Intensity> method, Proposed method maximum distance with the center of background (i.e. center with the highest cell score) from the each distance value of the nuclei center (i.e. center with the lowest cell score) gives a value that is closer to the target minima's, without losing too much information. Example preprocessed images can be seen at Figure 1-5. Also, an example for multi-stained image is at Figure 7-9.

Before starting the segmentation, a total variation denoising filter is applied to smooth the image while preserving the edges, along with a contrast limited histogram equalization algorithm.

2.2 Cluster Segmentation

Segmentation part consists of a series of K-means clustering and watershed algorithms. In the classical watershed algorithm, intensity plateaus are divided by a probabilistic or distance-based logic according to given seed points. This approach, however, has several downsides for

the cell segmentation. First of all, there can be more than one low-intensity seed points present for a large nucleus according to their neighbors, but in ideal conditions such a nucleus ought to be indivisible. Standard watershed algorithm divides this low intensity area to these multiple seeds, while the darkest one should have been allowed to dominate the region. Also, the methods to extract the seed points are generally parametrized, and very prone to over and under-segmentation. To prevent this, usual approach is the application of a region merging/splitting algorithm after the use of watershed, but the method we use removes the need of such measures.

First, the approach assumes that the darkest (lowest intensity) areas are closer to the target, which is a guaranteed information from preprocessing. Rest of the algorithm can be followed below:

- 1- Apply K-means clustering for given K (e.g. 10 or larger)
- 2- Assuming lighter areas are background and darker ones are nuclei, apply watershed algorithm K-1 times:
 - a. In first iteration, take darkest layer as seed, the preceding layer as grey area to be divided, and the rest as background.
 - b. In second iteration, take two darkest layer as seed, the preceding layer as grey area, and rest as the background.
 - c. Continue until one layer (sure background) is left
- 3- From the resulting masks, get the two darkest one. Compare them, apply watershed to the parts that intersects (taking the darker mask as seed) and take the non-intersecting parts of the lighter mask directly. Create a new mask with this method.

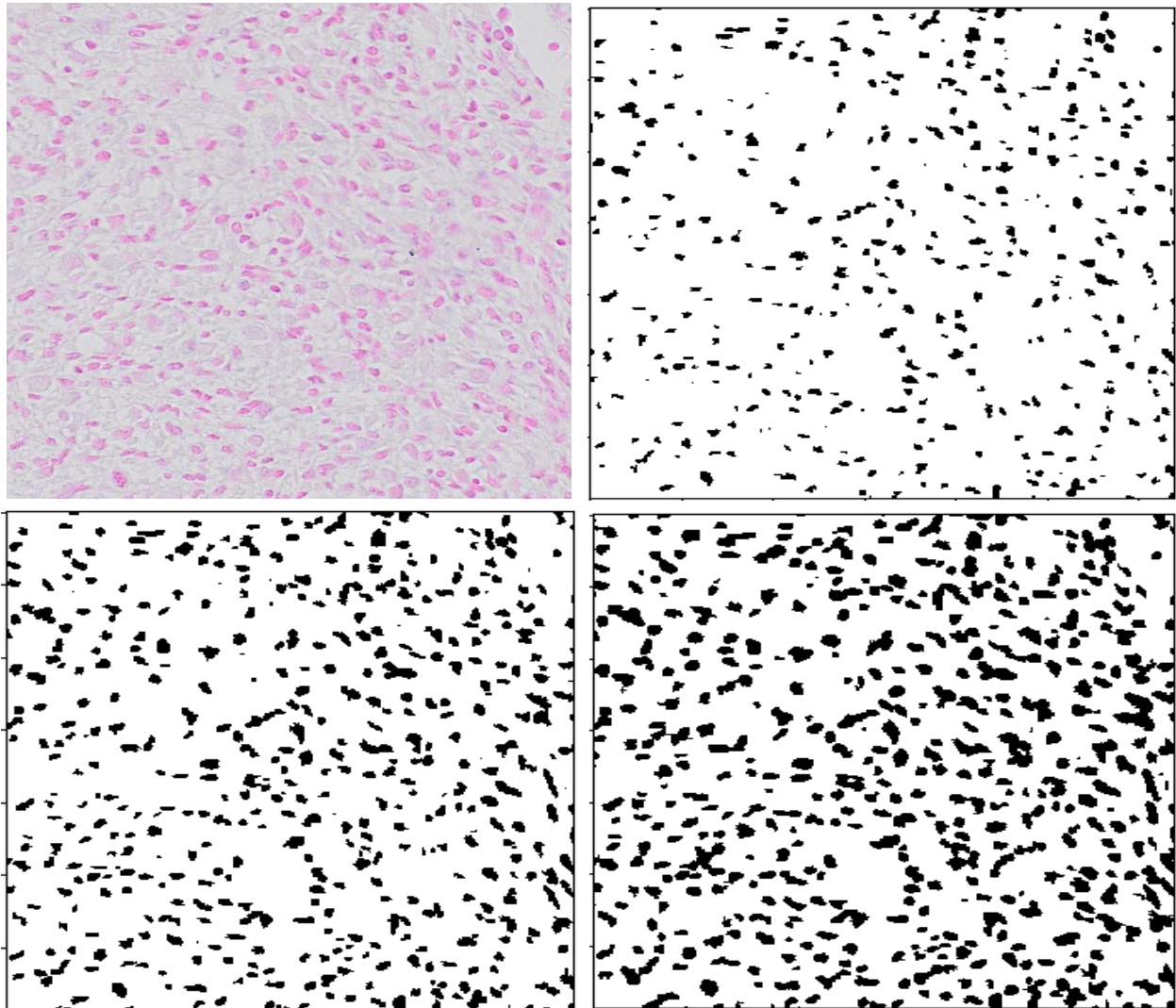


Figure 10-13 - From left to right, top to bottom: Initial Image, Mask 1 for $K=7$, Mask 2 for $K=7$, Mask 3 for $K=7$

- 4- Take the newly created mask from above, apply the algorithm to it with the third darkest mask, and continue as such iteratively. Meantime, store the average saturation of the cells and cell count.

- 5- Since we want both the average saturation and the cell count high, we get the geometric mean of them, and find the maximizing overlay and output it as the result of the segmentation.

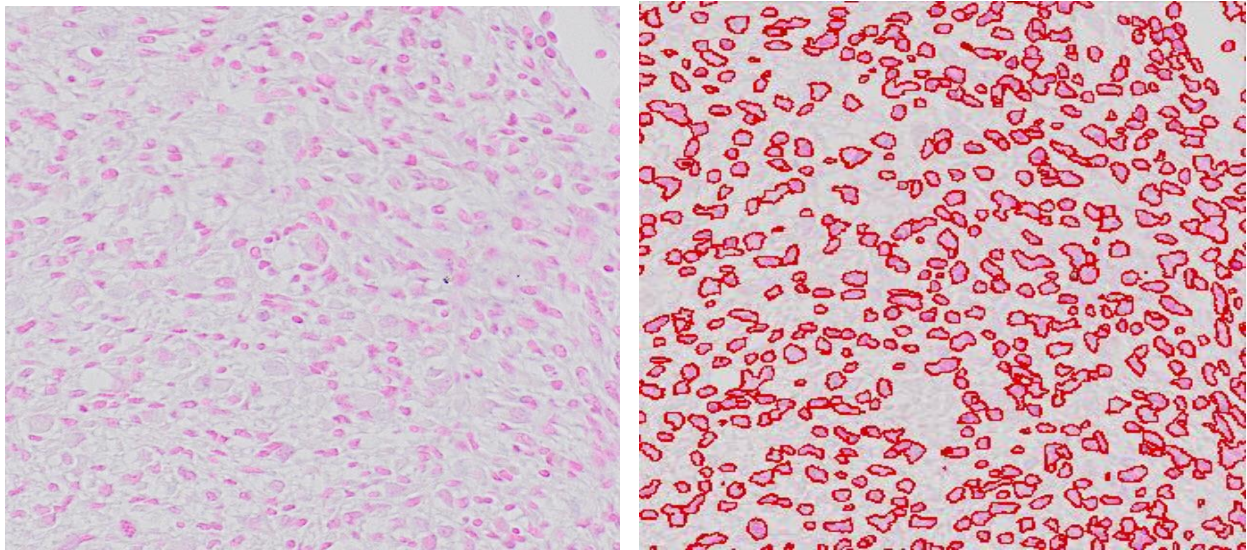


Figure 14-15 - From left to right: Initial image, Segmented Image for K=7 (recommended K is higher)

This approach also has an interesting result: Since the output cells can be classified according to their priorities which is supplied by the K-means algorithm, we have the data of sure nuclei gathered from our image, without the need of any supervision. This database can be used to extract meaningful features from the data in the future, and also utilized to create thresholds as in 2.3.

2.3 Cell Scoring and Quality Thresholding

As indicated in the previous chapter, the priority information for the output regions of the segmentation is accessible. To determine the regions that needs further processing, this dataset can be used. There is already a cell quality scoring system proposed at 2.1, and according to that formula a mapping of output clusters can be made.

To determine a threshold, regions that correspond to the nuclei from K=1 mask is extracted and their scores are calculated. At this point, we recommend the use of “trimmed arithmetic mean” which removes the highest and lowest 20% percent of the data to prevent manipulations, as used by ICE Libor to determine the interest rates that some of the world’s leading banks charge each other for short-term loans (Röman 33).

This arithmetic mean is the threshold which separates the regions that needs a further processing from the ones that are well-segmented. An example can be seen at the Figures 27-29.

2.4 Postprocessing and Cell Cutting

After the detection of the clusters that needs further processing, a morphological opening operator is applied to those regions with a disk kernel. This is utilized to get rid of the irregularities on region contours caused by watershed algorithm, and after rescoreing the image the count of labels that needs further processing drops down drastically. This effect can be observed at Figures 16-18.

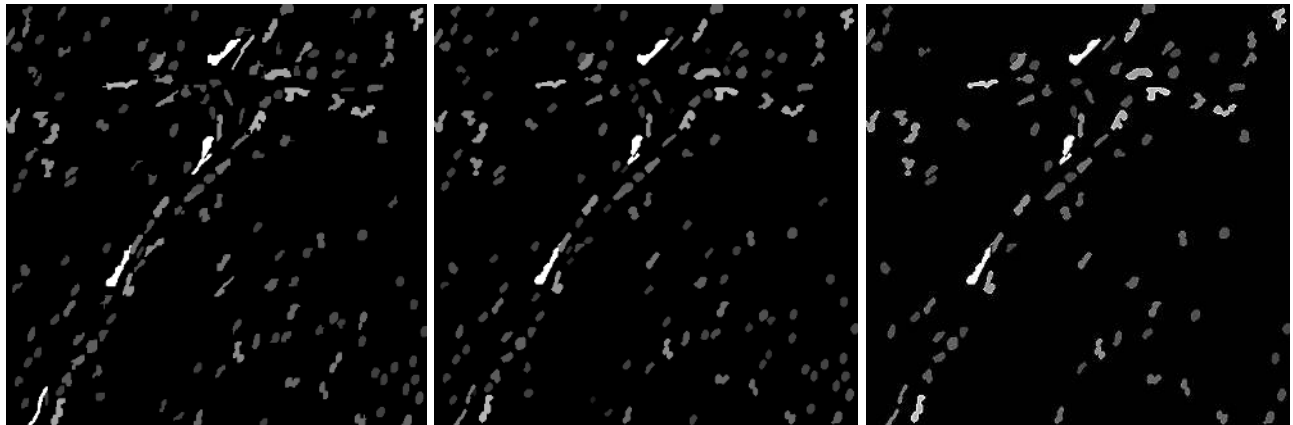


Figure 16-18 - From left to right: Scored image, Opened and rescored image, Remaining clusters

The areas that are still above the calculated quality threshold can be considered as multi-cell clusters or very deformed entities, therefore needs to undergo the cutting process.

Cell cutting is an extensively used algorithm for the nuclei that are hard to segment because they are too close, and makes use of the convexity defects of a cluster to draw cutting lines. For this, the approach described by Wienert et al. is used to determine a separation score for each possible line for the defects on a cluster. This score is calculated according to the angle of the cutting line and depth of respective defects (5-6).

The cutting process is applied recursively: After the detection of all convexity defects that are above a certain depth (e.g. 3 pixels), the separation score is calculated for each line and a cut is made on the line with least score. After the application of a morphological dilation with a very small kernel and the erasing of very small objects, the resulting regions are rescored and their mean is taken. If the resulting score is not larger than the initial score, and there are still possible cuts to make, algorithm continues the cutting process. The recursion stops when:

- Resulting regions have a higher score than the previous one,
- Both resulting regions are smaller than the possible cell size, or
- There are no convexity defects left.

For the former two cases, the recursion outputs the result preceding the final cutting, while for the last case result is the final output.

Chapter 3

Results and Discussion

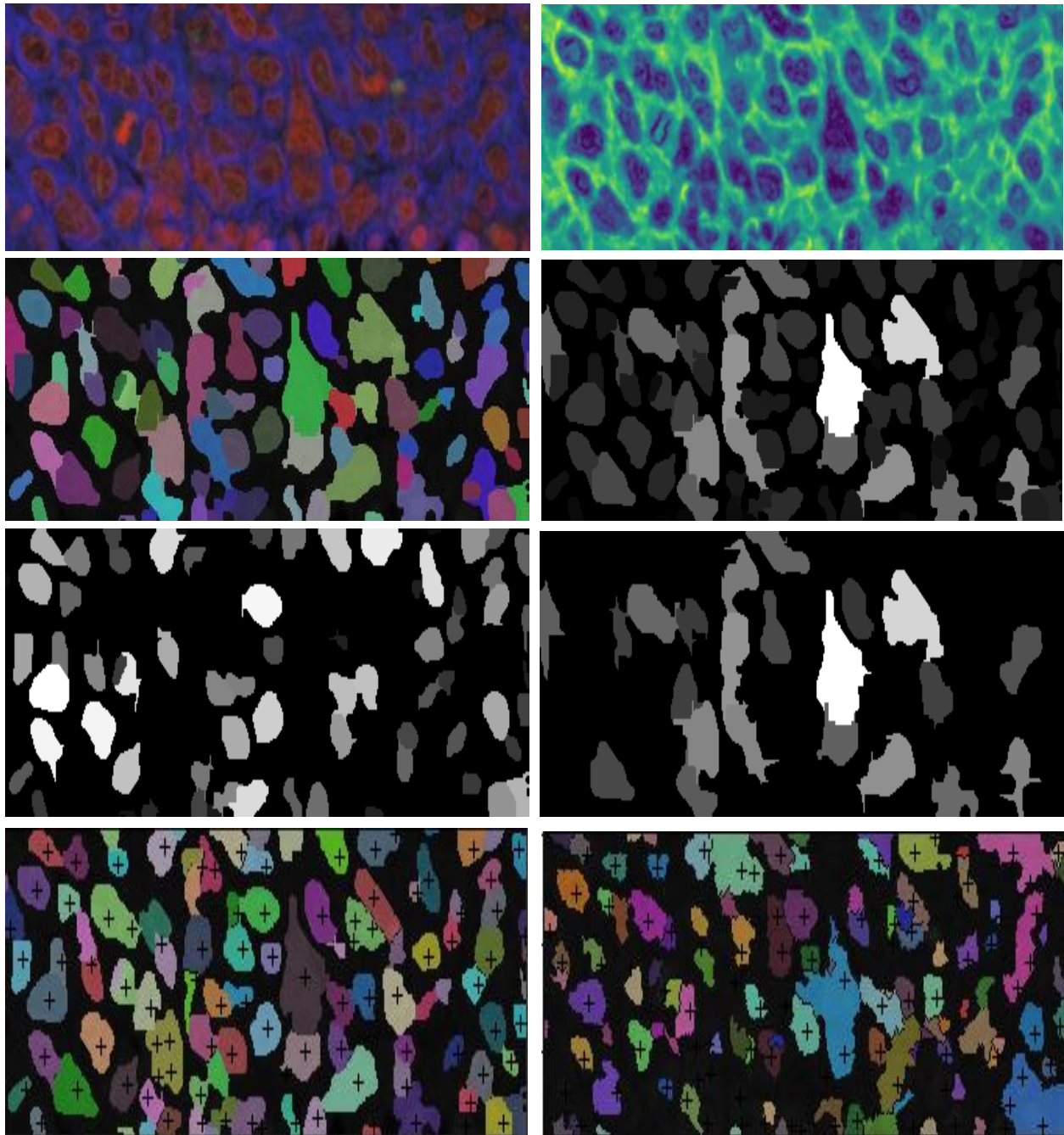
In order to create an acceptable output, two random tissue images with different staining techniques has been chosen. The methodology of extraction is applied to each image as described in the previous chapter. The chosen staining types included both unicolored (as in Figure 14 and 27) and multicolored (as in Figure 7 and 19) methods, which generally needs different approaches to extract the nuclei. The multicolored staining has also predetermined cell coordinates that will allow visual comparison with classical methods.

To compare with this newly proposed method, a standard watershed algorithm with Li binarization that creates a distance map to determine the sure background has been used. The results of the method and respective comparisons can be seen in the Figures 19-34.

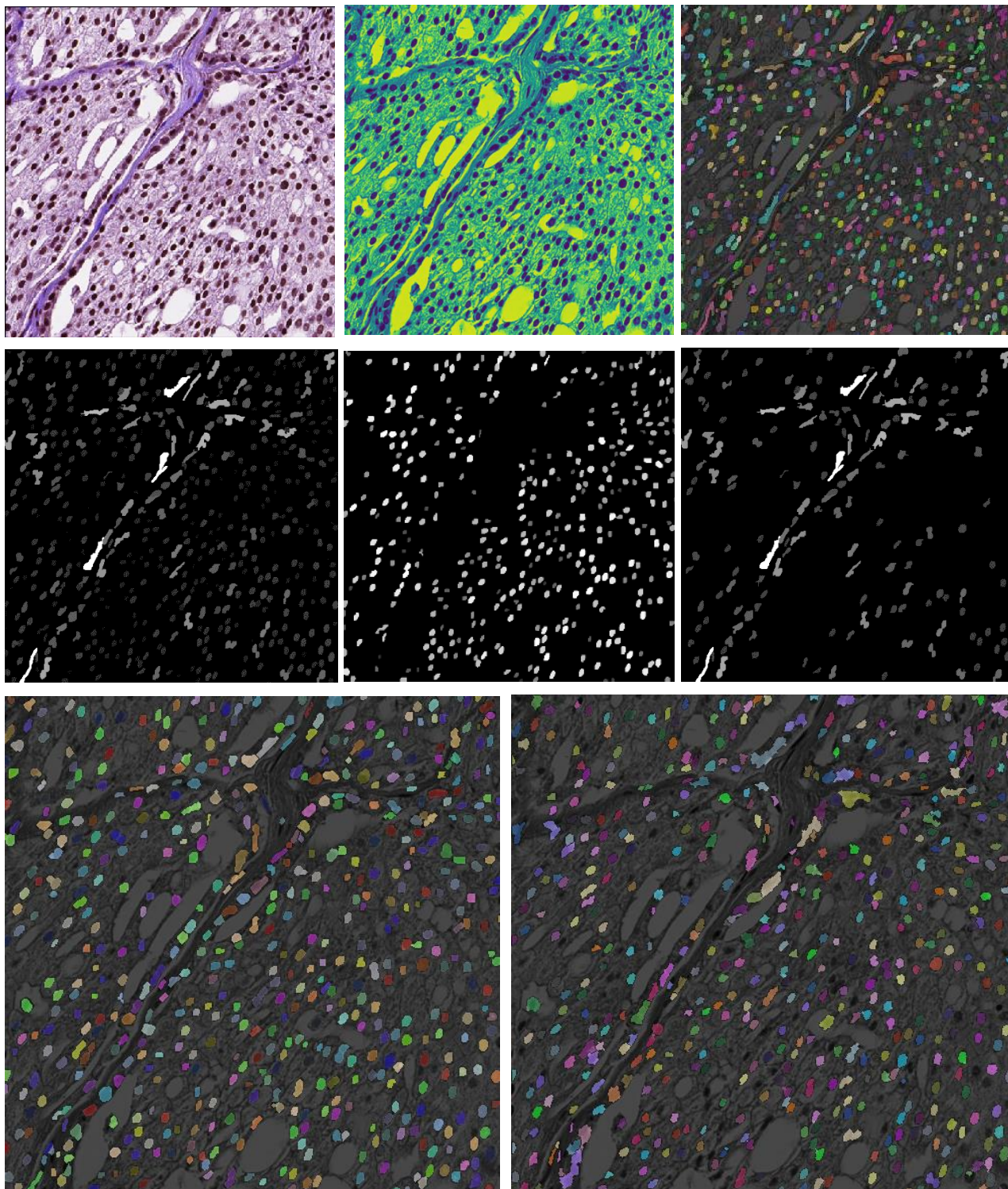
For the case with the unicolored image, we see a nearly perfect extraction completed by the proposed method at Figure 27. Even the nuclei that resides on cells with different composition which reacts adversely to the staining technique seems to be extracted with an acceptable quality. However, the result of the standard watershed segmentation lacks the detection of several nuclei even on high-contrast background, and on the part with different composition (with comparatively bluer background) the nuclei segmentation seems to fail.

Multicolored image shows a similar, but a more drastic result: Since the usual color to gray converting methods fail at these kinds of staining techniques, the classical watershed algorithm completely fails to create an acceptable segmentation. Proposed algorithm seems unable to

make the necessary post-segmentation at several places because of the proximity of nuclei, but the preprocessing and initial segmentation appears to serve the initial purpose.



Figures 19-26 - From left to right, top to bottom: Initial image from a part of Figure 7, Preprocessed image, Segmented image, Scored image, Cells below the threshold, Clusters above the threshold, Cut Image, State of the art watershed algorithm with distance mapping



Figures 27-34 - From left to right, top to bottom: Initial image, Preprocessed image, Scored image, Cells below the threshold, Clusters above the threshold, Result of the proposed method after cutting, State of the art watershed algorithm with distance mapping

Chapter 4

Conclusion and Future Work

4.1 Conclusion

In this report, a new and multi-purpose model has been proposed for cell and nuclei extraction from tissue staining images. At every step, either a new methodology is proposed, or a currently existing one is enhanced to fit the conditions that such a structure necessitates. As a result, close to ideal outputs are observed for several different types of tissue images, and better visual results are achieved comparing to simple K-means and watershed approaches. However, since there weren't any computational data for most of the tissue images at hand, the comparative studies lacked the necessary correct detection and segmentation ratios.

For the preprocessing part, the choice of the correct K has successfully worked for every sample image at hand, and normalization proved to be useful. However, some cases are observed at which this process is not enough, and a further augmentation of the center point is needed. In general, this step performs better than its state of the art counterparts with some aspects open to improvement.

Segmentation step is a fully functional alternative to a simple watershed and preserves any important information at hand, while providing a level of understanding on the quality of output cells. However, during the testing of the proposed method, a derivation is made from

standard watershed algorithm and “Power Watershed” method described by Couprie et al. is used. The writer considers this act as a mistake, since preprocessing seems to be more suitable to be used with standard watershed implementation.

Cell quality thresholding is a one-of-a-kind example that does not have a direct counterpart in the literature, as there isn’t a source of quality that is provided by the segmentation process. The thresholding appeared to be working without an obvious visual problem for the given dataset.

The final processing of cell cutting has some important flaws, since the disregard of the intensity values of the cell while making a cut is prone to create erroneous outputs. This approach needs to be developed with the inclusion of the information gathered through the intensity levels of the given region, thus preventing the division of a single nuclei.

4.2 Future Work

There exists several points in this report that can be improved by the information at hand. For example, cell cutting algorithm lacks the measures that segmentation algorithm uses to prevent cell nuclei divisions. As a result, some of the cuts are done through prioritized areas. This issue can be prevented by simple algorithms to penalize any such cuts using the information from the segmentation method, yet an alternative approach is needed in its place which may allow parabolic or hyperbolic lines to divide the aforementioned clusters.

Furthermore, the self-created data from the segmentation algorithm can be used to train supervised networks, which may allow automatic detection and division of badly segmented cells.

As a final note, the preprocessing method may be utilized for different areas of image processing, and needs to be tested for its performance on image analysis datasets.

Works Cited

- Bengtsson, E., et al. "Robust Cell Image Segmentation Methods." www.cb.uu.se/~carolina/web_ims/Bengtsson_PRIA_2004.pdf.
- Couprie, C, et al. "Power Watersheds: A New Image Segmentation Framework Extending Graph Cuts, Random Walker and Optimal Spanning Forest." 2009 IEEE 12th International Conference on Computer Vision, 2009, doi:10.1109/iccv.2009.5459284.
- Indhumathi, C., et al. "An Automatic Segmentation Algorithm for 3D Cell Cluster Splitting Using Volumetric Confocal Images." *Journal of Microscopy*, vol. 243, no. 1, Feb. 2011, pp. 60–76., doi:10.1111/j.1365-2818.2010.03482.x.
- Kanan, Christopher, and Garrison W. Cottrell. "Color-to-Grayscale: Does the Method Matter in Image Recognition?" *PLoS ONE*, vol. 7, no. 1, Oct. 2012, doi:10.1371/journal.pone.0029740.
- Kothari, Sonal, et al. "Automated Cell Counting and Cluster Segmentation Using Concavity Detection and Ellipse Fitting Techniques." 2009 IEEE International Symposium on Biomedical Imaging: From Nano to Macro, 2009, doi:10.1109/isbi.2009.5193169.
- Röman Jan R. M. *Analytical Finance: Volume II The Mathematics of Interest Rate Derivatives, Markets, Risk and Valuation*. Springer International Publishing, 2017.
- Meijering, Erik. *Cell Segmentation: 50 Years Down the Road*.
- Pinidiyaarachchi, Amalka. "Digital Image Analysis of Cells: Applications in 2D, 3D and Time." *DIVA, Acta Universitatis Upsaliensis*, 5 Feb. 2009.

Shah, Shishir. "Automatic Cell Image Segmentation Using a Shape-Classification Model." IAPR Conference on Machine Vision Application, 16 May 2008.

Wang, Weixing, and Hao Song. "Cell Cluster Image Segmentation on Form Analysis." Third International Conference on Natural Computation (ICNC 2007), 2007, doi:10.1109/icnc.2007.293.

Wienert, Stephan, et al. "Detection and Segmentation of Cell Nuclei in Virtual Microscopy Images: A Minimum-Model Approach." Scientific Reports, vol. 2, no. 1, Nov. 2012, doi:10.1038/srep00503.

Xiaoqiang, Ji, et al. "Cell Image Segmentation Based on an Improved Watershed Algorithm ." 8th International Congress on Image and Signal Processing, 2015.

Xing, Fuyong, and Lin Yang. "Robust Nucleus/Cell Detection and Segmentation in Digital Pathology and Microscopy Images: A Comprehensive Review." IEEE Reviews in Biomedical Engineering, vol. 9, 2016, pp. 234–263., doi:10.1109/rbme.2016.2515127.